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Grouped with:

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ENGG 151.01-A

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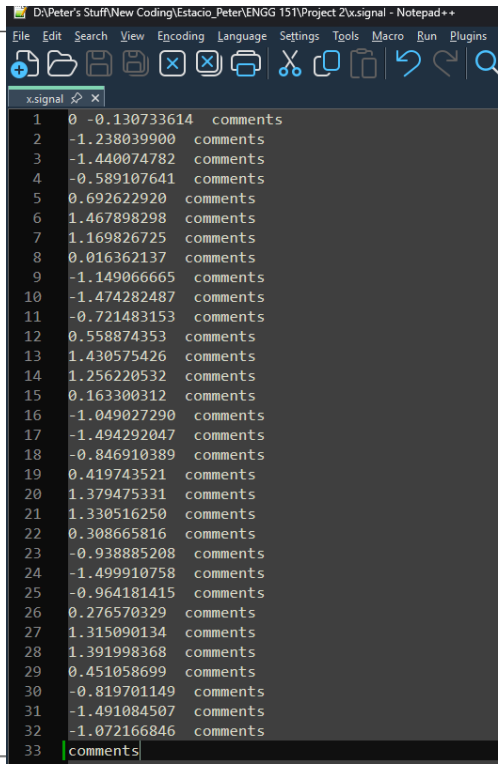
Self-Evaluation for Project 2: Fourier Transform

```
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 2> ./dft x.signal 32 4 8 8
Log file used: dftlog.txt
Signal file 'x.signal' with starting index 0 and duration 32 read.

Rectangular Results
==================================
Frequency (Hz)   Real Part       Imaginary Part
4.000000         -16.016414      -1.271327
4.500000         -2.091738       23.908673
5.000000         14.523652       1.379051
5.500000         0.000000        0.000000
6.000000         4.458294        0.479847
6.500000         0.000000        0.000000
7.000000         2.498667        0.291875
7.500000         0.000000        0.000000
8.000000         1.690067        0.206077

Polar Results
==================================
Frequency (Hz)   Magnitude       Phase (Degrees)
4.000000         16.066792       -175.461577
4.500000         24.000000       95.000000
5.000000         14.588977       5.424093
5.500000         0.000000        0.000000
6.000000         4.484042        6.143102
6.500000         0.000000        0.000000
7.000000         2.515656        6.662659
7.500000         0.000000        0.000000
8.000000         1.702584        6.951986
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 2>
```

sample screenshot – executable main run and displaying results of main program

x.signal

0	-0.130733614
	-1.238039900
	-1.440074782
	-0.589107641
	0.692622920
	1.467898298
	1.169826725
	0.016362137
	-1.149066665
	-1.474282487
	-0.721483153
	0.558874353
	1.430575426
	1.256220532
	0.163300312
	-1.049027290
	-1.494292047
	-0.846910389
	0.419743521
	1.379475331
	1.330516250
	0.308665816
	-0.938885208
	-1.499910758
	-0.964181415
	0.276570329
	1.315090134
	1.391998368
	0.451058699
	-0.819701149
	-1.491084507
	-1.072166846

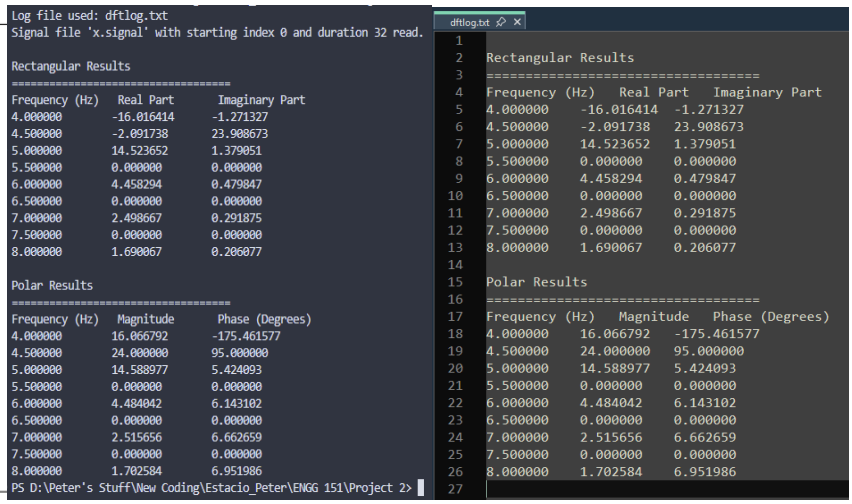


sample screenshots – comparison between basic test signal and input signal file used in program

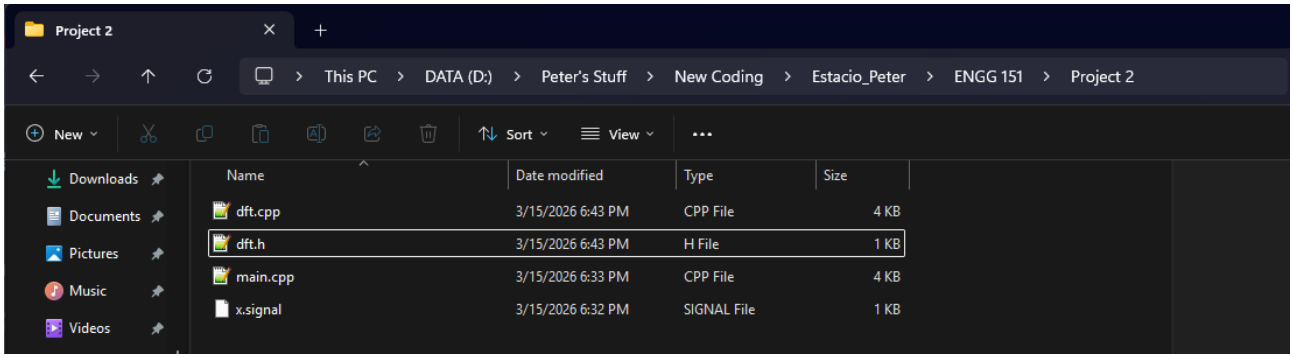
resulting DFT

frequency (Hz)	real part	imaginary part
4	-16.016414	-1.271327
4.5	-2.091738	23.908673
5	14.523652	1.379051
5.5	0.000000	0.000000
6	4.458294	0.479847
6.5	0.000000	0.000000
7	2.498667	0.291875
7.5	0.000000	0.000000
8	1.690067	0.206077

frequency (Hz)	magnitude	phase (degrees)
4	16.066792	-175.461577
4.5	24.000000	95.000000
5	14.588977	5.424093
5.5	0.000000	-62.499637
6	4.484042	6.143102
6.5	0.000000	-0.912811
7	2.515656	6.662659
7.5	0.000000	7.360520
8	1.702584	6.951986



sample screenshots – comparison between basic test results, console output, and program output file




```
* Executing task: C/C++: WINDOWS BUILD (g++.exe)

Starting build...
cmd /c chcp 65001>nul && C:\msys64\mingw64\bin\g++.exe -fdiagnostics-color=always -g *.cpp -o "D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 2\dft.exe"

Build finished successfully.
* Terminal will be reused by tasks, press any key to close it.
```

sample screenshot – compilation on Visual Studio Code with directory files

```
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 1\test> g++ --version
g++.exe (Rev8, Built by MSYS2 project) 15.2.0
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This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 1\test>
```
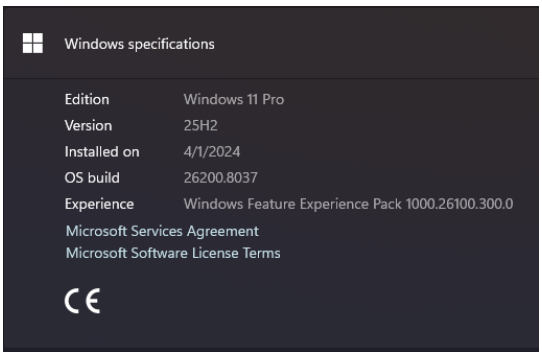
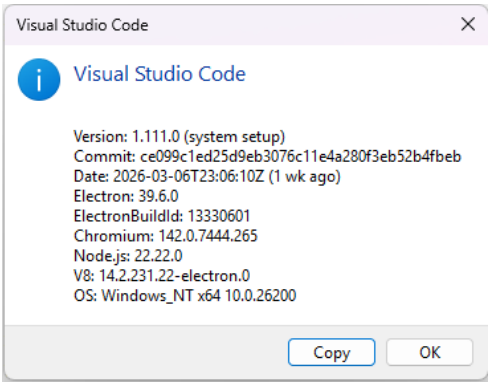

compiler version – g++ version 15.2.0 (Rev8, Built by MSYS2 project)



operating system - Windows 11 Pro, 25H2, build 26200.8037



IDE – Visual Studio Code ver 1.111.0

Code Locator:

code items	file name	line numbers
DFT function	dft.cpp	113-144
declaration of double * that are passed to the DFT function	main.cpp	43-47
separate processing of first line of signal file	dft.cpp	61-84
input validation of integers in processing signal files	dft.cpp	10-25
input validation of floating point numbers in processing signal files	dft.cpp	27-42

Project Evaluation - Fourier Transform		
Item	Points	Rubrics
early work	8/8	(all or nothing) 8: at most 10% of the code in the final implementation differs from that in early work submission
implementation and testing environment report	4/4	4 - all instructions followed correctly
basic test	50/50	50 - correctly computes the DFT of basic test data
	6/6	6 - screenshot submitted showing the correctly computed DFT of basic test data as it appears on the console output
generality of DFT	6/6	6 - the DFT is computed correctly and efficiently at all times
DFT function	6/6	6 - the DFT is computed correctly and efficiently at all times, by a function as specified
processing of signal files	4/4	4 - The file is opened only once and the first line of the signal file is processed correctly and separately from code processing the rest of the signal file
input validation	4/4	4 - integers and floating point numbers are fully validated when signal files are processed
command line	4/4	4 - application processes command line arguments correctly at all times
application feedback	4/4	4 - application produces required outputs and provides appropriate feedback at all times, as specified
self-evaluation	4/4	4 - self-evaluation accurate (or evaluating this item leads to an error)
total	100/100	